Research Statement

As the third precinct in Minneapolis burned the night of May 28th, 2020, a panoply of possible worlds seemed to spring forth from its embers. For many, it was a sign of impending lawlessness and the need for an even stronger police force, or at least the need to return to a more *civil* politic. For others, it brought forth visions of communities unplagued by the specter of state-sponsored violence and racial subjugation. In the following months, we would see the battle for these disparate futures play out not just on the streets, but also in the varied courts of public opinion – news media, social media, and every strata of government, from local to federal. At each turn, it seemed, policing advocates, journalists, and abolitionist organizers mobilized statistics and unique interpretations of policing data to identify the roots of what much of the public now acknowledged to be a pandemic of racialized police violence. For my own small part to play in this national debate, I helped analyze SFPD dispatch data to point to places where the department's budget could be easily cut and worked with the DefundSFPD campaign to share this analysis with local officials and other activist groups. Seeing these struggles over the perception of social reality play out at various scales, it only solidified my commitment to peeling back the veneer of "objectivity" and "neutrality" that currently surrounds everything *data-driven*.

To date, much of my research has attempted to peel back this veneer by centering the subjectivity of measurement and revealing the values embedded in the systems we attempt to feed data into. From interrogating the demographic categorizations employed in the pursuit of algorithmic fairness to inverting the machine learning pipeline to probe our practices of measurement (rather than taking them as unbiased lenses through which to view the world), my work has attempted to enable better alignment between datasets and algorithmic systems with the experiences of data subjects. With the training and support of the UC Berkeley I School, I am looking to extend this line of work through a grassroots approach to studying the collection of data that enables oppressive sociotechnical systems as well as potential methods for resisting these forms of surveillance and control.

Though I certainly have more to learn, my research and studies during my Undergraduate and Masters degrees and my time as a researcher at the Partnership on AI (PAI) have laid a strong foundation for me to do this type of work. Most recently, I have been leading a workstream looking into the risks of collecting demographic data to enable the production of "fairer" algorithmic systems. Briefly put, most proposed algorithmic fairness strategies coming out of academia and industry require practitioners to have access to data subjects' sensitive attributes, such as their race, gender, or sexuality. Last year I designed and conducted a semi-structured interview project with 38 participants who were working on issues of algorithmic bias in an assortment of industries, guiding conversations on what categories of demographic data are actually accessible, how they get used, and what types of concerns exist

around them. The academic paper stemming from this study was published at FAccT 2021 and is the writing sample for this application.

Perhaps unsurprisingly, we heard from most interviewees that they did not have access to most attributes of interest, prompting some members of the Partnership on AI community to ask how we might start collecting this data responsibly. This led me and my colleague, Sarah Villeneuve, to scope out a second phase of this work looking at the full range of risks from collecting these sensitive types of data, moving beyond the often debated privacy concerns. By critically analyzing academic and corporate documentation and drawing from a wide range of scholarship, we filled in the picture of how privatized demographic data collection, even under the guise of enabling fairness practices, can produce significant harm. As one of many possible examples, we identified ways that fairness techniques embed effectively colorblind definitions of fairness, such that collecting demographic data actually enables corporations to individualize responsibility for the impacts of systemic racism even as they say they are increasing the fairness of their systems. These types of misrepresentations of reality have already been shown to exist in policing data where demographic data collection has already been standardized, which brings into question how we might resist these misrepresentations going forward.

In a related workstream, my colleagues at PAI and I have been exploring alternative models of data governance for sensitive data. Earlier this year I helped design and run a CRAFT workshop at FAccT 2021 that used speculative fiction to get attendees thinking about the shortcomings of today's data governance strategies and infrastructures and start imagining alternatives that center autonomy and participation over more exploitative values.

Going back to my experiences as a Graduate and Undergraduate researcher, I spent much of my time more closely engaging with technical systems, learning and building upon various methods of Critical Technical Practice. In one project with a longtime collaborator, Tom Gilbert, we looked at how Machine Learning could be made more "fair" by using it as a tool for interrogating unjust measurement practices instead of as a mechanism for prediction. (This work was published at AIES 2020, under the title "Towards a Just Theory of Measurement.") For my undergraduate thesis I critiqued my own research project that I conducted as an undergraduate intern at the Center for Human-Compatible AI at UC Berkeley. By unpacking the values and assumptions implicitly embedded in the "human-compatible" AI I was tasked with developing efficient algorithms for, I went on to show how the project's initial framing guaranteed societal incompatibility from the outset.

One key research question I would love to explore during my PhD is how to support policing and prison abolitionist groups in creating and managing datasets that can help characterize policing and carceral abuses and inefficiencies while simultaneously resisting

surveillance and datafication efforts from above. As Crooks and Currie[1] suggest, the goal of grassroots data collection cannot be to simply prove the moral bankruptcy of the current state of affairs – this is already known and experienced by entire communities, and those with the power to change it already have all the data that would be needed as evidence. Rather, I would like to elaborate methods for how community-led data creation can be used to make an impassioned case for community organizations, advocacy groups, and other local groups to get involved in the struggle against policing. Building off of a wealth of prior scholarship in similar domains, we would take as a starting point for this work that collecting data from systemically marginalized individuals is often done to responsibilize them for systemic issues[2] or advance a deficit narrative of their community[3] and frequently leaves them feeling exhausted, used,[4] and jaded,[5] all of which points to the importance of rooting our research questions in the work already being done on the ground. Recent methodological developments in participatory action research such as Sucheta Ghoshal's "Grassroots Praxis of Technology"[6] and Miriam Asad's "Prefigurative Design"[7] provide crucial touchpoints for this work, as they tackle the question of how to ensure that technologies embedded in resistance and movement building mirror the values held by grassroots organizers.

The Berkeley I School has an immense amount of expertise in relevant domains that would all be very instructive for this proposed research area. There is Professor Niloufar Salehi's experience with crafting technologies for social movements, exploring notions of restorative justice in virtual communities, and interrogating the interplay between power and data, Professor Deirdre Mulligan's experience with rigorously operationalizing values in technology and exposing those that went previously unexamined, and Professor Jenna Burrell's experience with ethnographic studies of adaptive technology (non-)use and agency over algorithmic systems. Having already taken courses with Professors Mulligan and Burrell (as well as a partial course with Professor Salehi), I would love the opportunity to train under them again as I work towards becoming an engaged scholar, educator, and (hopefully) professor myself.

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[1] Crooks, Roderic, and Morgan Currie. "Numbers will not save us: Agonistic data practices." *The Information Society* (2021): 1-19.
[2] Eubanks, Virginia. *Automating Inequality: How High-Tech Tools Profile, Police, and Punish the Poor*. St. Martin's Press, 2018.
[3] Walter, Maggie. *Delivering Indigenous Data Sovereignty*. https://www.youtube.com/watch?v=NCsCZJ8ugPA. 2020 Fay Gale Lecture, Academy of the Social Sciences in Australia.
[4] Petty, Tawana, et al. "Reclaiming our data: interim report, Detroit." (2018).
[5] Barabas, Chelsea, et al. "Studying up: reorienting the study of algorithmic fairness around issues of power." *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*. 2020.
[6] Ghoshal, Sucheta. A Grassroots Praxis of Technology: View from the South. Diss. Georgia Institute of Technology, 2020.
[7] Asad, Mariam. Sculpting Reality from our Dreams: Prefigurative Design for Civic Engagement. Diss. Georgia Institute of Technology, 2019.